

Clovis SFEIR

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Profile

IMT Atlantique engineering student specialising in performance-critical systems and quantitative software for markets and industrial infrastructure. Built five C++20 systems benchmarked against industry references or verified against labeled ground-truth: ITCH 5.0 feed handler at **97 ns p99** (**8.2×** Aquis Exchange), options engine at **302×** QuantLib, an Ornstein-Uhlenbeck-driven crude-spread trading engine, and an IEC-104 ICS intrusion detector with **100% attack recall**.

Education

IMT Atlantique – Engineering Degree in Software Engineering (Apprenticeship) *Sep 2025 – Jun 2028*
Nantes, France *Algorithms & Data Structures, OS, Computer Networks, Distributed Systems, Cloud Computing, AI*
Bachelor in Computer Science *Sep 2022 – Jun 2025*
La Rochelle, France *Class rank: 1/85 | Software Engineering, Databases, Cybersecurity*
HRT Macro Placement Challenge 2026 (Hudson River Trading/Partcl) – **24th/124** (top 20%); hybrid analytical placer: force-directed global placement with timing-aware legalization; proxy cost **1.095** beats SA baseline (2.125) and RePLace (1.458); field includes Stanford, Princeton, ETH Zürich.

Projects

ITCH 5.0 Market Data Feed Handler *C++20, AF_XDP, AVX2, lock-free*

- AF_XDP/BPF steers ITCH 5.0 multicast to a hugepage UMEM ring (kernel bypass); busy-poll receiver on an isolated core, SCHED_FIFO prio 99; AVX2 `_mm256_cmpeq_epi64` level search; open-addressed order map ($\leq 50\%$ load).
- Add **97 ns**, cancel **13 ns** (**8.2×** Aquis Exchange); standard CI: 315/46 ns; sustained **15.7 M msgs/s** at p99=128 ns over 10 s; 0 dropped packets; zero allocations.

Options Pricing Engine

C++20, AVX2, SABR, Crank-Nicolson FD

- AVX2 Black-Scholes (4-wide `_mm256d`), minimax normal CDF avoids `libm erf()`; Greeks in one pass; **150.9 M prices/s** (**302×** QuantLib); SABR surface (Hagan 2002 + Antonov), L-M calibration, 5×9 surface in 174 μ s.
- American options via Crank-Nicolson FD (sinh-grid, Thomas TDMA): 110 μ s/option; Sobol quasi-MC (antithetic + CV + IS); 10 M-config validation vs QuantLib, max error $< 10^{-6}$.

Crude Oil Spread Trading Engine

C++20, stochastic processes, order book

- Models refining margins (3:2:1 crack spread, generalized N:M:K) and mean-reverting inter-crude spreads via an Ornstein-Uhlenbeck process (Euler-Maruyama) with a rolling z-score signal, matched by a price-time-priority order book.
- OU stationary distribution verified against theory (500K-step sim); crack-spread formula checked to 10^{-9} ; order-book submit at **720 ns** (non-crossing)/**1.24 μ s** (crossing); 58 assertions/21 tests, ASan+UBSan clean.

Reservoir Diffusivity PDE Solver

C++20, OpenMP, finite-difference

- Explicit FTCS and implicit Crank-Nicolson finite-difference schemes for the reservoir diffusivity (pressure-transient) equation; verified via grid-refinement convergence study against the closed-form `erfc` analytical solution – error ratio converges to **4.0** as dx halves (textbook 2nd-order accuracy) across 5 refinements.
- OpenMP-parallelized stencil/assembly loops; measured (not assumed) scaling identifies the Amdahl's-law-bounded sequential Thomas solve and memory-bandwidth-bound behavior below L3 size; 13 test cases, 40,300 assertions; Docker image runs the full suite at build time.

ICS Intrusion Detection (IEC 60870-5-104)

C++20, protocol parsing, explainable rules

- Rule-based detector for IEC 60870-5-104 (no auth/encryption by protocol design) – the actual telecontrol protocol behind the 2016 Kyiv blackout – mapped to real attack patterns: control commands to unexpected breaker objects (the exact method used), sequence-number gaps (replay/desync), u-format/command floods.
- Parses APCI+ASDU at **39.5 ns** (**25.3 M msg/s**); end-to-end parse+detect 337 ns; **100% recall, 0 false positives** on a labeled synthetic trace (100 records, 15 attacks across 5 rules); 153 assertions/37 tests, ASan+UBSan clean.

Work Experience

Infotel

AI Engineer Apprentice – Lab IA

Nantes, France

Sep 2025 – Jun 2028

- Built a Rust-based agentic orchestration prototype (specs $\rightarrow$ blueprints $\rightarrow$ vertical slices $\rightarrow$ artifacts $\rightarrow$ tests $\rightarrow$ security reports); ownership model prevents use-after-free in concurrent task graphs; quality gates separate build/test/security scans from artifact-only checks; 928 Rust tests; incremental Merkle indexing for $O(\text{changed-files})$ re-index with requirement $\rightarrow$ code $\rightarrow$ test traceability.

Excelia Group

Software Developer Intern / AI Software Engineer Intern

La Rochelle, France

Two internships: Jan–Mar 2025 & May–Jul 2024

- RAG pipeline (chunking, OpenAI embeddings, flat FAISS, cross-encoder re-ranking): **40% reduction** in lookup time vs. keyword-search; cut p95 latency 30–40% via embedding cache + asyncio parallelisation; Prometheus/Grafana, Docker, GCP.
- Real-time interview simulator (Next.js/React, GPT-4o, ElevenLabs TTS): streaming responses, scenario-branching state machine, A/B prompt-scoring harness.

Technical Skills

Languages

C++20, Python, Rust, TypeScript, SQL

Perf / Systems

AF_XDP, SIMD/AVX2, lock-free SPSC, OpenMP, cache-aware layout, SCHED_FIFO, acquire/release atomics, ICS/SCADA protocol parsing (Modbus, IEC-104)

ML / Quant

Black-Scholes/SABR, Sobol quasi-MC, Kalman filter, Ornstein-Uhlenbeck, finite-difference PDE, Johansen cointegration, LightGBM/XGB/CatBoost, FAISS

Infrastructure

Docker, Prometheus/Grafana, GitHub Actions CI, GCP, Redis, ZeroMQ

Frameworks

Flask, Node.js/Express, React/Next.js, REST/gRPC